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Thermal expansion behavior of carbon fiber reinforced chemical-vapor-infiltrated silicon carbide composites from room temperature to 1400 °C

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Abstract

Thermal expansion behavior of C/SiC composites with three different preform architectures from room temperature (RT) to 1400 °C was investigated. It was found that they related to changes of interfacial thermal stress and microstructure due to interaction of fibers and matrix. The longitudinal coefficients of thermal expansion (CTE) of 1D and 3D C/SiC composites and in-plane CTE of 2D C/SiC composites changed in such similar way that increasing linearly with increasing temperature, reaching maximum values at about 900 °C, and fluctuating from 900 to 1400 °C. However, they had different increasing rates below 900 °C. In addition, transverse CTEs of 1D and 3D C/SiC increased linearly and both changed in a similar way as bulk CVD SiC material at low temperature range. And then, they increased rapidly from 1200 to 1400 °C after a decrease at 900 °C.

Keywords: C/SiC composites; Thermal expansion; Interfacial thermal stress; Preform architectures

1. Introduction

Carbon fiber reinforced silicon carbide (C/SiC) composites are promising structural materials in fields of aeronautics and astronautics. Under the condition of unstable high temperature, interfacial thermal stress is difficult to avoid due to thermal mismatch of three compositions, i.e. carbon fiber, PyC interphase and SiC matrix. And it often causes changes of mechanical properties and microstructure of the composites [1–3]. Thermal expansion behavior of C/SiC composites depends on interfacial thermal stress [4]. It can be known that how interfacial thermal stress in C/SiC composites changes by investigating thermal expansion behavior of the composites. Therefore, a systematic study on thermal expansion behavior of C/SiC composites is needed to understand the evolution of interfacial thermal stress and microstructure during their service. In this paper, CTEs of C/SiC composites with three different preform architectures from RT to 1400 °C were investigated. Although 1D C/SiC composites are seldom used, the study on its thermal expansion behavior is helpful to understand those of other two kinds of C/SiC composites because interfacial thermal stress in one-dimensional (1D) C/SiC composites is relatively simple.

2. Experimental

2.1. Fabrication of C/SiC composite specimens

Three kinds of fiber preforms were prepared: one-dimensional(1D) preform by putting 3 K T-300 carbon fibers in order, two-dimensional (2D) by laminating 1 K T-300 woven carbon fabrics, and three-dimensional (3D) by braiding 3 K T-300 carbon fibers in four-step method. The preforms were infiltrated with both PyC interphase and SiC matrix to fabricate C/SiC composites by low pressure chemical vapor infiltration method (LPCVI) using butane and methyltrichlorosilane (MTS) as the reactive materials [5]. The infiltration conditions of PyC interlayer were as follow: temperature 960 °C, pressure 5 KPa, time 20 h, Ar flow 200 ml·min^{-1}, butane flow 15 ml·min^{-1}. The infiltration conditions of SiC matrix were as follow: temperature

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1000 °C, pressure 5 KPa, time 120 h, H_2 flow 350 ml·min^{-1}, Ar flow 350 ml·min^{-1}, and the molar ratio of H_2 and MTS 10.

Specimens were machined from the fabricated composites and further deposited with SiC under the same conditions to prepare a SiC coating and make the composites denser. Additionally, to investigate the controlling factors of thermal expansion behavior of C/SiC composites, a bulk SiC material was prepared by chemical vapor deposition method (CVD) [5] in the following deposition conditions: temperature 1100 °C, time 200 h, H_2 flow 400 ml·min^{-1}, Ar flow 400 ml·min^{-1}, and the molar ratio of H_2 and MTS 3–5.

2.2. CTE measurements

DIL 402C dilatometer made by NETZSCH Company was employed for measurements of longitudinal and transverse CTEs of 1D and 3D C/SiC composites, as well as in-plane CTE of 2D C/SiC composites. The required specimen size was $3.5\times3.5\times20$ mm^3. All measurements were conducted in nitrogen atmosphere from RT to 1400 °C.

3. Results and discussion

3.1. Longitudinal CTEs of 1D and 3D C/SiC composites

Relationship between longitudinal CTEs of 1D and 3D C/SiC composites and temperature was illustrated in Fig. 1, as well as in-plane CTE of 2D C/SiC composites which would be discussed in the next section. As shown in Fig. 1, longitudinal CTEs of 1D and 3D C/SiC composites changed in such similar way that increasing linearly with increasing temperature below 900 °C, reaching maximum values at about 900 °C, and fluctuating in the temperature range of 900 to 1400 °C, in that the braiding angle of 3D C/SiC composites was just 22 ° and distribution of interfacial thermal stress in both of C/SiC composites could be considered approximatively as identical.

Tensile stress and microcracks were produced in SiC matrix and compressive stress in carbon fibers during the cooling of 1D and 3D C/SiC composites from manufacture temperature to RT because the longitudinal CTE of carbon fibers was much lower than that of SiC

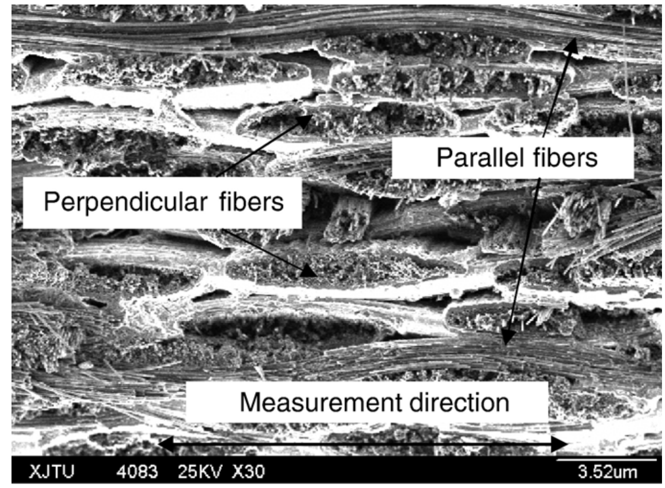

Fig. 2. A SEM photograph of weaving architecture of 2D C/SiC composites.

matrix. With the temperature increasing, SiC matrix expanded and stresses in C/SiC composites were released gradually. Due to the interstices in matrix, the longitudinal CTEs of both composites were lower than the bulk CVD SiC material. Therefore, CTE of SiC matrix played a leading role and longitudinal CTEs of both composites followed law of the matrix from RT to 900 °C. Thermal expansion of matrix was restricted by carbon fibers as the temperature up to 900 °C, and then CTEs of both composites decreased at near 1000 °C. At one time, the transition of compressive stress in carbon fibers and tensile stress in SiC matrix happened. Some carbon fibers were fractured with the increasing tensile stress in fibers, which induced that the expansion of SiC matrix was not restricted gradually. Therefore, CTEs of both composites increased again from 1100 to 1200 °C, and were equal to that of CVD SiC at 1200 °C. CTEs of both composites decreased sharply from 1200 to 1300 °C and were holding a constant level in the following temperature range because stresses in C/SiC were redistributed and expansion of the matrix was restricted again after fibers were fractured as well as negative longitudinal expansion of C/PyC substrate happening as the temperature was above 1200 °C [6]. Therefore, Longitudinal CTEs of both composites were primarily controlled by that of carbon fibers and interfacial thermal stress from 900 to 1400 °C. Compared to that of 1D C/SiC composites,

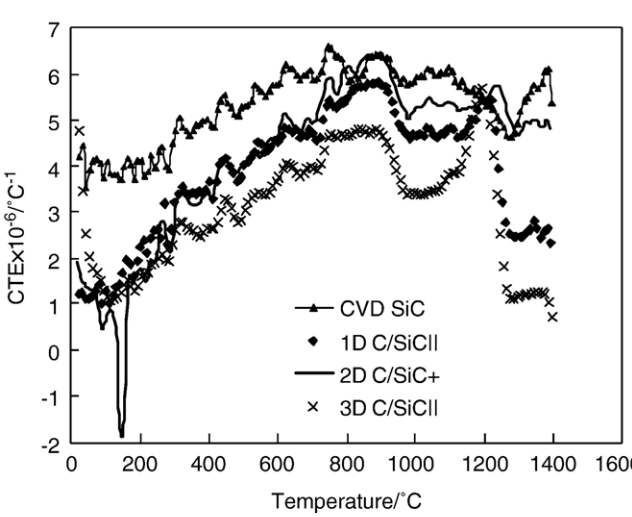

Fig. 1. Longitudinal CTEs of 1D and 3D C/SiC composites and in-plane CTE of 2D C/SiC composites from RT to 1400 °C.

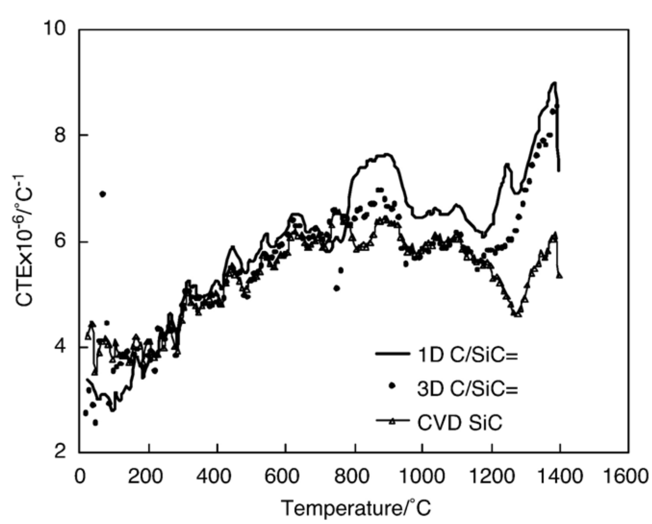

Fig. 3. Transverse CTEs of 1D and 3D C/SiC composites from RT to 1400 °C.

Table 1
Controlling factors of thermal expansion of C/SiC composites with different preform architectures

Composites		At low temperatures (<900 °C)	At high temperatures (>900 °C)
1D C/SiC	Longitudinal ‖	SiC matrix	C fiber and interfacial thermal stress
	Transverse =	SiC matrix	C fiber and interfacial thermal stress
2D C/SiC	In-plane +	SiC matrix and radial expansion of perpendicular fibers	Axial expansion of parallel fibers and interfacial thermal stress
3D C/SiC	Longitudinal ‖	SiC matrix	C fiber and interfacial thermal stress
	Transverse =	SiC matrix	C fiber and interfacial thermal stress

longitudinal CTE of 3D C/SiC composites was lower owing to the larger restriction of fibers on expansion of SiC matrix derived from its better preform architecture and higher loading capacity of fibers.

3.2. In-plane CTE of 2D C/SiC composites

As shown in Fig. 1, in-plane CTE of 2D C/SiC composites was larger than longitudinal CTEs of other two composites and changed in a similar way as them in the whole temperature range, while had a higher increasing rate below 900 °C.

For 2D C/SiC composites, half of fibers were in parallel direction to the observed surface and the other half were in perpendicular direction (Fig. 2). The radial expansion of perpendicular fibers was larger than that of SiC matrix and could not be ignored. Tensile stress and cracks vertical to axis of the parallel fibers were produced in SiC matrix when the composites were cooling from manufacture temperature to RT. Meanwhile, interstices were generated between the perpendicular fibers and matrix caused by shrinkage of the former. When the temperature increased again, thermal stresses were released owing to the radial expansion of the perpendicular fibers and SiC matrix. Interstices and matrix cracks offered room for this expansion which was also restricted by the parallel fibers. Therefore, the radial expansion of perpendicular fibers became a controlling factor besides CTE of SiC matrix in thermal expansion behavior of 2D C/SiC composites, which caused a higher increasing rate below 900 °C. It was also the reason that in-plane CTE of 2D C/SiC composites was larger than those of other two kinds of C/SiC composites in the whole temperature range. However, it was lower than that of bulk CVD SiC due to the existence of thermal stress. In-plane CTE of 2D C/SiC composites fluctuated above 900 °C and then intersected that of CVD SiC at 1200 °C. Matrix cracks vertical to the axis of parallel fibers were healed mostly, and tensile stress was generated in parallel fibers, which inhibited radial thermal expansion of perpendicular fibers and that of SiC matrix. Simultaneously, negative longitudinal CTE of C/PyC substrate took place [6]. So, in-plane CTE of 2D C/SiC composites decreased gradually in the temperature range of 900∼1300 °C. SiC matrix was cracked in direction of axis, which released radial thermal expansion of perpendicular fibers at 1300 °C, and then CTE of the composites increased again. However, fracture of parallel fibers stopped above 1200 °C where CTE of 2D C/SiC composites reached a peak value. Consequently, in-plane CTE of 2D C/SiC composites was controlled by expansion of parallel fibers in axial direction and interfacial thermal stress from 900 to 1400 °C.

3.3. Transverse CTEs of 1D and 3D C/SiC composites

Relationship between transverse CTEs of 1D and 3D C/SiC composites and temperature was illustrated in Fig. 3. Tensile stress in fibers and compressive stress in SiC matrix were produced and interfacial debonding happened partly when C/SiC composites were cooling from manufacture temperature to RT because radial CTE of C fibers was much higher than that of SiC matrix. Residual stresses were released and expansion increased with increasing temperature. Transverse CTEs of both composites had the same behavior as bulk CVD SiC from RT to 900 °C, and they had just a little higher value than the latter due to porous structure of high modulus SiC, although radial CTE of fibers was pretty high. Compressive stress was produced in fibers and transverse CTEs of both composites decreased with the temperature increasing. At 1200 °C, SiC matrix cracked in axial direction of fibers and then the expansion of fibers was released. Transverse CTEs of both composites increased rapidly above 1200 °C. Therefore, they were controlled by radial CTE of C fibers and interfacial thermal stress at high temperature range. Transverse CTE of 3D C/SiC composites was lower than that of 1D C/SiC composites because radial thermal expansion of C fibers was restrained greatly and interfacial thermal stress was complicated in 3D C/SiC composites.

4. Conclusions

CTEs of 1D and 3D C/SiC composites changed in a similar regulation due to the similar distribution of interfacial thermal stress and the identical controlling factors in the same temperature ranges, controlled by CTE of SiC matrix at low temperatures and by CTE of C fibers and interfacial thermal stress at high temperatures. Different behaviors were observed in the investigation of in-plane CTE of 2D C/SiC composites because radial expansion of perpendicular fibers became a controlling factor besides CTE of SiC matrix in thermal expansion behavior of 2D C/SiC composites.

Controlling factors of thermal expansion of C/SiC composites with different preform architectures were summarized in Table 1.

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